Setting Up and Running VASP on H100 GPU

1 Building VASP from Source

1.1 Install NVIDIA HPC SDK

```
wget https://developer.download.nvidia.com/hpc-sdk/22.11/
    nvhpc_2022_2211_Linux_x86_64_cuda_multi.tar.gz
tar -xvf nvhpc_2022_2211_Linux_x86_64_cuda_multi.tar.gz
cd nvhpc_2022_2211_Linux_x86_64_cuda_multi
./install
```

1.2 Configure Environment

- Add your NVHPC module path to the `$MODULEPATH` environment variable in your `.bashrc`.
- Reload your environment:

  ```
  source ~/.bashrc
  ```

- Check module availability:

  ```
  module avail nvhpc
  ```

1.3 Allocate H100 GPU

- Request a node with an H100 GPU using your cluster's resource allocation command (e.g., `salloc`, `sbatch`, or equivalent).

1.4 Load Required Modules

```
module purge
module load nvhpc/22.11
module load FFTW.MPI/3.3.10-gompi-2023b
```

1.5 Build VASP

- Navigate to your VASP source directory.
- Copy the makefile:

```
cp arch/makefile.include.nvhpc_acc ./makefile.include
```

- Edit `makefile.include`:
  - Set `-gpu=cc90` for `CC`, `FC`, and `FCL`.
  - Ensure the CUDA version is set to 11.8.
- Set environment variables:

```
export PATH=/path/to/nvhpc/comm_libs/mpi/bin:$PATH
export LD_LIBRARY_PATH=/path/to/nvhpc/comm_libs/mpi/lib:
    $LD_LIBRARY_PATH
```

  (Replace `/path/to/nvhpc` with your NVHPC directory path.)
- Compile VASP and run the tests:

```
make DEPS=1 all
make test
```

2 Using VASP as an End User

2.1 Allocate H100 GPU

- Request an H100 GPU node using your cluster's resource allocation command (e.g., `salloc`, `sbatch`, or equivalent).

2.2 Load the Module

```
module load gpu_h100/vasp/6.5.0
```

2.3 Prepare and Run VASP

- Prepare your input files: `INCAR`, `POSCAR`, `POTCAR`, and `KPOINTS`.
- Create a test directory and move into it:

```
mkdir vasp_test
cd vasp_test/
```

- Run VASP using `mpirun`, choosing the appropriate executable (`vasp_std`, `vasp_gam`, or `vasp_ncl`) based on your calculation needs:

```
mpirun -np 1 ./vasp_std
```

Notes

- The executable names vary depending on the type of calculation:
  - `vasp_std`: standard VASP.
  - `vasp_gam`: for Gamma-point-only calculations.
  - `vasp_ncl`: for non-collinear magnetic calculations.